

Team RAVIR: A Unified Approach for Vessel Segmentation from Retinal Scans

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Abstract

Retinal Images consist of many modalities and give us insight into many pathological conditions and vascular conditions. Automatic segmentation of vascular structure is an active research direction and there are many datasets available for this task. But this research direction suffers from the scarcity of large datasets that hampers training SOTA deep learning models. Researchers have explored transfer learning-based and domain adaptation approaches to resolve this problem. Through this work, we present a novel IR retinal vessel segmentation dataset with 25 annotated samples and through rigorous experimentation using 3 very different deep learning based models, establish that we can circumvent transfer learning and domain adaptation-based approaches while doing vessel segmentation on retinal scans. We present our results and a novel dataset.

Keywords: Retinal Scans, Semantic Segmentation, Deep Learning, Domain Adaptation, Transfer Learning

1. Introduction

The vasculature of the retina is directly visible through the eye, allowing for images to be acquired from multiple imaging modalities. Captured images can provide important information about the cardiovascular system([Diaz-Pinto et al. \(2022\)](#)) and aid in assessing cardiovascular conditions including stroke, coronary heart disease, and dementia. The vasculature of the retina has also been associated with systemic diseases such as hypertension([Stanton et al. \(1995\)](#)), stroke([Wu et al. \(2017\)](#)), diabetes, and neurodegenerative disorders.

It has been hypothesized that changing retinal vascular structure can be associated with the early stages of diseases([Ikram et al. \(2013\)](#)). To quantify the state of the vasculature first, there need to be sufficient information to make comparisons with, and second, the vascular structure needs to be mapped and identified as arterial or venous. This process is time-consuming because a professional is needed to annotate the images and manually perform the segmentation.

With the advent of deep learning we have seen many downstream tasks in the image domain, of which image segmentation is a prominent one. Many different approaches have been taken to tackle image segmentation tasks. The task has also been defined in different categories: Semantic segmentaion, Instance segmentaion and Panoptic segmentaiton

(De Geus et al. (2018)). Different approaches have been proposed over the years, where the traditional ones were most pre-processing heavy and the latter ones being more end-to-end. The field has seen many autoencoder based approaches(Ibtehaz and Rahman (2020)) to transformer based(Strudel et al. (2021)) ones and now diffusion(Ho et al. (2020)) based models are also claiming superiority for different benchmark tasks. The recentmost advances are taking a more zero shot and promptable turn like the Segment Anything Model by microsoft(Kirillov et al. (2023)). But still the field requires quality datasets for evaluation and benchmarking of these models. The introduction of deep learning into the field of medical images allows for the semantic segmentation of images to be much less time-consuming in the future. Medical image segmentation comes with its own challenges as this are high-stake tasks and medical decisions may be affected due to the performance of these networks(Haque and Neubert (2020)).

In this work we

- Explore three approaches to the semantic segmentation of retinal images across imaging modalities and classes of segmentation.
- Present a novel dataset RAVIR+ with 25 annotated IR retinal scan samples for artery and vein segmentation.
- Show that domain mismatch from different imaging modalities does not impact the model performance in vessel segmentation task.

2. Retinal Image Scanning Modalities and Challenges

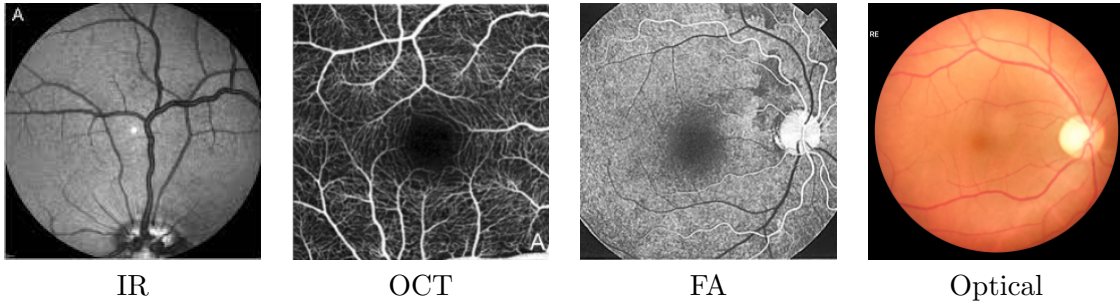
There are many types of retinal imaging modalities each with its own advantages and disadvantages. Infrared Reflectance (IR) images use infrared wavelengths of light to penetrate the tissue and the reflectance is measured. This reflectance is then visualized allowing for allowing the deeper retinal layers to be observed. IR imaging, however, provides limited information on blood flow and lacks detail compared to other modalities.

Optical Coherence Tomography Angiography (OCTA) images utilize low-coherence near-infrared light to capture images within the retina. These images are very high resolution and are able to capture volumetric blood flow information. OCTA is used for the diagnosis of retinal and vascular disease including diabetic retinopathy, and macular degeneration. However, OCTA images have limitations in imaging deeper vascular layers.

Fluorescein Angiography (FA) images are produced by injecting a fluorescent dye (fluorescein) into the bloodstream, then capturing an image of the retina while there is dye passing through the vasculature. FA images provide information on blood flow, leakage, and pooling of fluid in the retinal vessels, which allows these images to be used in the diagnosis and monitoring of retinal vascular diseases. However, FA is invasive, carries risk with the necessary injection, and may not provide as much detail as OCTA.

Color photographs or optical imaging are widely used for screening and monitoring due to their relative simplicity and wide availability. The major disadvantage of optical images is the limited information on blood flow and the lack of penetration into deeper retinal layers. Optical images are only able to capture the surface level of the retina.

Figure 1: Popular modalities of retinal scans



The data from the different modalities pose potential domain mismatch in distribution and hence data from different modality have been treated as a OOD(out of distribution) in literature([Hatamizadeh et al. \(2022\)](#)). Representative images are shown in Figure 1.

3. Datasets

3.1. Existing Datasets

Table 01: Existing Datasets

Dataset	Training images	Designated Test Split	Classes	Modality
RAVIR	26	20	3	IR
IOSTAR	30	-	3	Color
CHASE DB1	20	8	2	Color
HRF	45	-	2	Color
STARE	20	-	2	Color
RAVIR+(Proposed)	25	-	3	IR.

Table 01 displays the existing datasets containing retinal segmentation masks. RAVIR and IOSTAR datasets were utilized in the first test with three class segmentation because they all have corresponding masks. For the experiment where two class segmentation occurred all 5 existing datasets were used.

The IOSTAR dataset was acquired by a scanning laser ophthalmoscope (SLO) from patients suffering from diabetic retinopathy. This dataset consist of 30 color images.

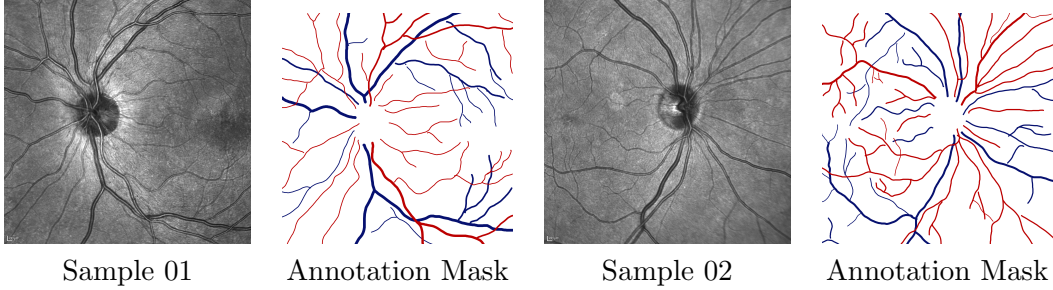
CHASE DB1 dataset includes retinal images from a hand-held fundus camera with 14 images both left and right eye for a total of 28. 20 of these images are used in the training set with the remaining 8 being used as a test set.

The HRF dataset contain 45 color images, of the 45 images 15 are from patients with diabetic retinopathy, 15 are from patients with glaucoma, and 15 are from healthy patients.

3.2. Our Dataset: RAVIR+

We developed an annotated retinal vessel segmentation dataset with 25 IR scans. We used Adobe photoshop and a digital drawing setup to annotate the samples and all the samples

Figure 2: Annotated samples with masks in proposed RAVIR+ dataset



have been annotated by medical professionals. A sample for the segmentation is provided below. For convenience, we call this dataset RAVIR+ onwards. In Figure 2 we show a couple of samples from our RAVIR+ dataset. All the IR image samples have been collected from publicly available online sources.

4. Experimentation Design

4.1. Experiment 01: Artery-Vein Segmentation

In this experimentation, we consider the segmentation task as a 3 class segmentation where the segmentation objective is to segment the two types of blood vessels (veins and arteries) separately.

Of the datasets available, only IOSTar, RAVIR and RAVIR+ have 3 class annotations. So we develop a 70-30 split from each of the datasets and create the extended train-test split comprising samples from all the 3 datasets. Upon going through the dataset, we obtained 7.33% Vein Area, and 5.45% Artery Area.

4.2. Experiment 02: Blood Vessel Segmentation

In this experimentation, we performed a 2 class segmentation where the objective was only to segment blood vessels. No distinction between the vessel category is taken into consideration. This line of experimentation have been done because the majority of the datasets in this research direction do not contain separate segmentation masks for vessels and arteries.

In this experimentation, all the datasets have been compiled and a 70-30 split have been done on all the datasets and compilation is done, so that the training and test data contains equal feature diversity from each of the different modalities. Also, while compiling the datasets, the 3 datasets from experimentation 01 have been converted to a 2 class segmentation dataset collapsing the two segmentation masks to maintain uniformity.

4.3. Models

We used 3 models to experiment the performance of deep learning models in the automatic segmentation of retinal blood vessels. We chose very different benchmark models to capture the feature diversity and the difficulty of the task.

- **UNET with a ResNet background:** UNET (Ronneberger et al. (2015)) is a benchmark model popular for semantic segmentation task. In our experimentation we are using an UNET with a ResNet-18 backbone, based on previous successes in using UNETs with a backbone in successful feature extraction in doing segmentation tasks.
- **Deeplab V3:** Deeplab v3 (Yurtkulu et al. (2019); Chen et al. (2017)) is an atrous convolution based segmentation model. This model has been shown to provide good results for image segmentation tasks.
- **MedSegDiff:** This is a diffusion model based segmentation model (Wu et al. (2022)). The forward process takes in the annotated masks and gets a noisy counterpart by iteratively adding Gaussian noise. The denoising process involves an UNet and also guidance from actual retinal scan embeddings.

While doing the experimentation, the UNET models have been trained for 400 epochs and the deeplabv3 models for 100. These models have been trained on Google Colab/Kaggle. The diffusion model based experimentation have been carried out on lambdalabs cloud server (1xA100 GPU) for 100k steps. While training the networks,

4.4. Evaluation

We report the Dice score (Bertels et al. (2019)) of segmentation for the vessel segmentation classes, as it is a standard metric for biomedical image segmentation task evaluation.

The Dice score, also known as the Dice Similarity Coefficient, is a measure of the similarity between two sets of data, usually represented as binary arrays. In the context of image segmentation, for example, the Dice score can be used to evaluate the similarity between a predicted segmentation mask and the ground truth segmentation mask for each of the segmentation classes. Dice score of 0 indicates no overlap hence bad segmentation. A dice score of 1 indicates perfect segmentation that follows the ground truth.

The Dice score is calculated as follows:

$$Dice\ score = \frac{2 \times |A \cap B|}{|A| + |B|} \quad (1)$$

5. Results

5.1. Results for Experimentation 01: Artery and Vein

As mentioned before, in experimentation 01 we performed 3 class segmentation using a compiled version of the 3 datasets containing granular data. From Table 5.1 we can see that the performance for the vein segmentation is consistently better compared to the

Table 02: Segmentation Performance for Artery-Vein Segmentation. The dice score for artery segmentation and vein segmentation are shown for the 3 different models.

Model	Artery Dice	Vein Dice	Avg Dice Score
UNET w/ Resnet Backbone	0.78	0.80	0.79
Deeplab V3	0.75	0.75	0.75
MedSegDiff	0.76	0.78	0.77

Table 03: Segmentation Performance for Blood Vessel Segmentation

Model	Avg Dice Score
UNET w/ Resnet Backbone	0.87
Deeplab V3	0.94
MedSegDiff	0.91

performance for the artery segmentation, across different models. We also observe that the simplest UNET based model performed best overall leading to a 0.79 dice score. One factor that might be responsible for the underperforming of the high complexity models can be lack of training data. As the dataset is very small, the larger models did not converge and generalize well, compared to the comparatively smaller model.

5.2. Results for Experimentation 02: Blood Vessel Segmentation

In experimentation 02, we performed 2 class segmentation using a compiled and modified version of all the datasets in performing blood vessel segmentation. In this experiment we only predict if a pixel belongs to a blood vessel of the background. No artery-vein distinction is maintained.

As expected, the average dice score is increased for the same models as the number of classes have decreased. Also, here we see improved performance for the deeplab model compared to the other ones that we suggest is due to the increased amount of training data present. Although medsegdiff is the SOTA model for medical image segmentation ([Wu et al. \(2022\)](#)), we see it underperforming due to lack of more data.

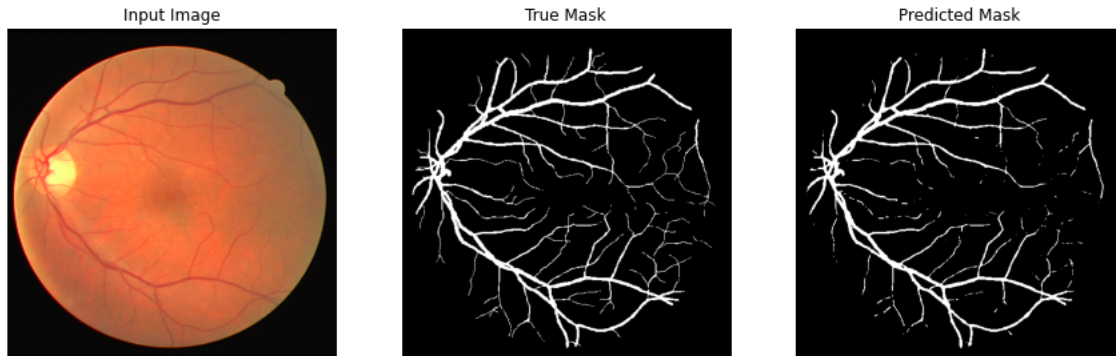
For both the experiments, we observe that although the training and test dataset contain samples from different modalities but small in number, the model learned necessary features despite having data from different distributions and learned the task well.

In Figure 03, we present the segmentation performance for a representative sample.

6. Conclusion and Future Work

In this work, we present a unified approach to segment vessels from retinal scans. We present a new dataset and perform semantic segmentation on 2-class and 3-class variations of retinal vessel segmentation task from retinal scans of different modalities. We report performance of 3 very different types of models and found the simpler model perform better, that we think is due to the scarcity of data for this task. We leave accumulation and vessel annotation of more IR retinal scan data as a future task.

Figure 3: Representative prediction from UNET based model on 2 class segmentation.



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